ClearSight

Deciding how much to discount end-of-season stock — and when.

An AI-assisted markdown copilot for multi-store seasonal retailers

MBA ZG583 · Management of AI Products · Capstone

THE PROBLEM

One national discount. Three hundred different stores.

At end of season, planners decide markdown depth and timing in spreadsheets, at national level. Analysing thousands of items across hundreds of stores by hand is not feasible — so one schedule is applied to all of them.

1 Cut too early

Margin given away on stock that would have sold closer to full price.

2 Cut too late

Stock clears at or below cost, having eaten store and warehouse space.

Markdown is one of the largest controllable margin levers in seasonal retail — and this decision repeats for every seasonal item, every season.

THE USER

Priya has been burned in both directions

Seasonal Merchandise Planner

- 3–4 categories
- ~2,000 seasonal items a year
- 300+ stores
- Runs exit planning in Excel
- Answers to a trading manager for every markdown taken

What she needs

A defensible reason for each decision, and the ability to say no to the tool.

What loses her

One confidently wrong recommendation she followed and had to explain upward.

Why that matters

Error cost is asymmetric. Over-discounting a winner loses margin now; holding a loser clogs stores. The design has to respect both.

THE AI CASE

Why not just a better rule?

Flat markdown schedule	Assumes stores behave alike. Weather, demographics and local competition say otherwise.
Sell-through rules	Rules need history. Private-label seasonal ranges are substantially new each year.
Dashboards / BI	Show what happened. Cannot estimate what happens under a 20% cut now vs 35% in three weeks.
More analysts	The volume is the problem. People do not scale, and do not make the decision consistent.

The AI task: predict residual sell-through per store cluster at candidate price points — generalising from product attributes where direct history does not exist.

THE PRODUCT

It recommends. A person decides.

Rank items by risk → Forecast by cluster → Recommend depth + timing → Show confidence → Planner approves or overrides

No automatic repricing. Below a confidence threshold, ClearSight shows no recommendation at all — and says why.

MVP — deliberately narrow

- One category, one exit window, one market
- Ranked worklist, projected sell-through, confidence band
- Approve / adjust / reject with a reason code

Product, not project

- Core: forecasting, ranking, confidence routing, override capture
- Customisation: category taxonomy, store clusters, integration
- Fixed cost spreads across customers, not carried by the first

WHAT THE PLANNER SEES

Confident — and not confident

← **Kids' rash vest, 5 colours**

2,400 units · $186k at stake

RECOMMENDATION

Cut 25% — this week

Projected 82% sell-through by end of window
vs 61% under the current flat schedule

CONFIDENCE

HIGH

14 analogue lines · 3 prior seasons

Why this recommendation

- Sell-through is 18pts behind the same week last season
- 9 of 12 clusters are tracking below plan; coastal is not
- Analogue lines cleared at 25% with minimal residual
- Holding 2 more weeks projects 71% — below the 80% target

Largest driver: cluster-level sell-through gap

Options compared

Hold — no cut	61%
Cut 15% now	74%
Cut 25% now	**82%**
Cut 35% now	88%

Projected sell-through · 35% cut recovers
less margin despite higher volume

Approve | Adjust | Reject

Adjust and Reject require a reason

Reason (appears on Adjust / Reject)

Competitor activity · Range being repeated next season · Store feedback · Stock quality issue · Other

A recommendation, its confidence, and the alternatives

The planner can approve, adjust, or reject — each one click

← **Cooler bag 24L**

1,100 units · $97k at stake

NO RECOMMENDATION

Routed to manual decision

ClearSight does not have enough signal to recommend a markdown for this item.
The decision stays with you.

Why confidence is low

- New line — no prior-season history for this item
- Only 2 analogue lines matched, both weakly
- Cluster demand signals conflict: 6 up, 5 down

Confidence would improve with 3+ more weeks of
in-season sales, or a stronger attribute match.

What you can still see

- Sell-through to date by cluster
- Stock on hand and cover weeks
- Comparable lines and how they cleared

Data without a recommendation
attached to it.

Record my decision

Logged as a manual decision — outcome still tracked for evaluation

The same screen when the model lacks signal

No recommendation at all — and it says why. The decision stays with her, with the evidence still shown.

This second screen is the product.

DATA & EVALUATION

Beat the current rule — or stop

The AI task

Forecasting + recommendation: residual sell-through per store cluster at candidate price points, generalised from product attributes and analogue lines.

The baseline

The retailer's own flat markdown schedule, plus last-season analogue matching. If the model cannot beat both on held-out seasons, it does not ship.

Technical	Forecast error by cluster · calibration of the confidence signal
Business	Margin recovered vs baseline · terminal stock share
Trust	Override rate and reasons · high-confidence recommendations that went wrong

Capstone data: synthetic or public multi-store retail data only. No confidential employer data — workflow knowledge is context, not input.

ECONOMICS

Priced on margin recovered — not per recommendation

Annual platform fee

Per banner, plus a per-category fee — sized as a clear fraction of conservatively estimated margin recovered.

Not per recommendation

Usage billing creates cost anxiety — and would suppress exactly the careful review behaviour the product depends on.

Cost shape

Variable	Inference and data pipeline — scales with catalogue size
Fixed per customer	Store clustering setup, integration, planner onboarding

RISK & GOVERNANCE

What could go wrong — and what stops it

Confidently wrong at scale	A high-confidence recommendation the planner trusted and should not have. The failure mode that ends the product.	Confidence thresholds · mandatory manual routing below them · no automatic price execution
Rubber-stamping	If every recommendation is accepted, the human-in-the-loop control is nominal.	Acceptance rate monitored as a health signal with a ceiling — not maximised · reason capture on every override
Feedback loops	After season one, the model observes outcomes of prices it influenced. Naive retraining bakes in its own errors.	Holdout categories retained · drift monitoring from pilot onwards

Launch blocker: if expected loss from high-confidence wrong recommendations exceeds expected gain from the rest, it stays in pilot.

RECOMMENDATION

Prototype — then a hard gate before anything else

Phase 1

Backtest against historical seasons. No interface.

GATE · Beat the flat rule — or stop here.

Phase 2

Planner MVP in shadow mode. Decisions not applied.

GATE · Planners engage and give real override reasons.

Phase 3

Live pilot, one category, matched control group.

GATE · Margin outcome vs control. No high-value failures.

A backtest that cannot beat a flat schedule ends the project. That is a finding, not a failure.

Where I would value feedback

Is the store-cluster wedge narrow enough for one capstone? · Does outcome-anchored pricing hold up without live retailer data?